

IEEE VR 2023 Workshops

Workshop: 3D Reconstruction, Digital Twinning, and Simulation for Virtual Experiences (ReDigiTS 2023)

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ABSTRACT:

Today, Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR) technologies are becoming fundamental across a wide variety of application domains. All these technologies fall under the Extended Reality (XR) spectrum. Areas in which XR technologies and, in particular, immersive experiences have been successfully introduced are expanding daily, from traditional areas, such as entertainment and video game production, to novel scenarios, such as industry, healthcare, smart cities, and autonomous vehicles, to mention a few. The spread of XR and VR is also driven by technological advancements that are making new ways of operating within virtual experiences easier, and sometimes even possible. In particular, the availability of new hardware and software tools make the use of “3D reconstructions”, “digital twins” and “simulations”

commonplace in many XR and VR applications. For example, the possibility to leverage networks with higher bandwidth than never before and lower latency makes it possible to transmit the amount of data needed for massive, real-time digital twins, whereas high-performance hardware enables the reconstruction of high-fidelity digital objects and environments, while lowering the time needed for high-demanding simulations. The aim of the *3D Reconstruction, Digital Twinning, and Simulation for Virtual Experiences* (ReDigiTS) workshop is to bring together researchers, practitioners, educators, and students to share ideas and promote further work in this growing area of 3D reconstruction, digital twinning, and simulation in VR and related technologies.

MESSAGE FROM THE REDIGITS WORKSHOP ORGANIZERS:

We are delighted to welcome you to the second edition of the *3D Reconstruction, Digital Twinning, and Simulation for Virtual Experiences* (ReDigiTS) 2023 workshop. The workshop is organized in conjunction with the *30th IEEE Conference on Virtual Reality and 3D User Interfaces* (IEEE VR 2023), Shanghai, China, March 25-29, 2023. We are proud to bring you yet another year of this workshop confirming that interest in these topics is continuing to grow. We would like to thank all those whose hard work made this edition of the ReDigiTS workshop possible. Special thanks to the reviewers who spend their time to ensure valuable reviews during a very short

review period. We want to express our sincere thanks to all the authors for their outstanding submissions without which this conference could not occur. Finally, we are grateful to the IEEE VR 2023 workshop chairs for allowing us to organize the second edition of the workshop and for their guidance throughout the process. We believe that the participants will find the ReDigiTS program fascinating, inspiring, and plan to contribute to future editions of this workshop.

Please enjoy the ReDigiTS 2023 Workshop!

OBJECTIVES:

The main objective of this workshop is to bring together researchers, practitioners, educators, and students to spur discussion and share ideas. For this reason, the workshop aims to attract a collection of high-quality submissions reporting research activities targeted to the next generation of immersive experiences, reporting the latest ideas, methodologies, applications, evaluations, case studies, prototype implementations, preliminary results and demos concerning 3D

reconstruction, digital twinning, and simulation in VR and related technologies. The target audience is characterized by researchers, students and practitioners who are interested in the design, development, and validation of 3D reconstruction-, digital twin-, and simulation-based virtual experiences in the broad fields of, e.g., industry, healthcare, cultural heritage, smart cities, education, etc.

ABOUT THE WORKSHOP:

This year the workshop received submissions from six countries, i.e., United States, Republic of Korea, Denmark, Italy, United Kingdom and Cyprus. The workshop collected submissions that can be categorized as full research papers, work-in-progress, and position papers. Each submission has been reviewed by at least two domain experts. Reviewers were invited to evaluate both the quality and the relevance of the paper to the workshop scope.

At the end of the reviewing process, six papers were selected. The final program includes the following papers.

In the paper titled “Reconstruction of Human Body Pose and Appearance Using Body-Worn IMUs and a Nearby Camera View for Collaborative Egocentric Telepresence” the authors present an approach capable of capturing and reconstructing a human pose and appearance over time by merging the information gathered by inertial measurement units (IMU) and AR headset. The results demonstrated improvements in the quality of body reconstructions with respect to approaches that envisage the use of either IMUs or AR cameras alone.

In the paper titled “Automated Multimodal Data Capture for Photorealistic Construction Progress Monitoring in Virtual Reality” the authors investigate the use of a framework for autonomous multimodal data collection and visualization in VR with the goal of supporting human decision-making activities.

The authors of “Towards Outdoor Collaborative Mixed Reality: Lessons Learned from a Prototype System” investigate challenges related to collaborative mixed reality (CMR) systems by presenting a prototype that allows remote VR users to collaborate with a local outdoor AR user. The VR users are immersed in a large-scale 3D model of the AR user’s physical space. The study highlights key areas requiring additional research to overcome the challenges of outdoor CMR systems.

The user study reported in the paper “Get a Variable Grip: A Comparison of Three Gripping Techniques for Controller-Based Virtual Reality” compared three techniques to grasp virtual objects. The techniques envisaged different behaviors and effects of gravity on virtual objects in response to a grasp operation. Results indicate improvements in terms of perceived realism for one of the techniques.

The authors of “A Study on Affordable Manipulation in Virtual Reality Simulations: Hand-Tracking versus Controller-Based Interaction” show the results of a user study aimed to compare the use of controllers with respect to optical hand-tracking for interacting with virtual environments. Results indicate that the hand-tracking system enables more realistic interactions. These interactions were also perceived as more consistent with the real counterpart.

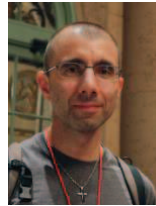
Finally, the authors of “Simulating Location-Based Experiences in VR” present joint research with Aardman Animations studio whereby a user study was carried out to analyze the effect of distortion in a large-scale projection-based VR installation. Results confirmed the benefits of using VR to investigate how the design choices can affect distortion and quality of the experience.

These papers demonstrate how vibrant this area of research actually is, and we believe they will contribute to raising further awareness to this growing area of research.

This year we also have the pleasure to include a keynote speaker. More specifically, Prof. Manuela Chessa (University of Genoa, Italy) shares her visions on challenges that concern the interactions within Extended Reality (XR) environments in the talk entitled “Interacting in Extended Reality: perception and action from real to virtual”.



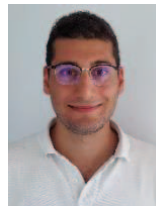
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